Thread Line Record: Model A-T — Adversarial Audit of the Angular Cancellation Programme

Governance Record Prepared by Model A-T

Purpose of this document

This document records the complete audit engagement between Model A-T and the human operator during an adversarial review of the Angular Cancellation manuscript and its Coq formalisation. It is written as a governance record rather than a performance summary.

The governing standard that eventually emerged during the audit was:

> For each chunk under review, Model A-T was to give only one of two verdicts: **mathematical error found** or **no mathematical error found**. If an error was claimed, Model A-T had to identify the exact broken inference.

Model A-T did not begin the exercise operating at that standard. Early in the audit, Model A-T repeatedly generated manufactured objections, misread expository prose as load-bearing mathematics, conflated formal interfaces with hidden gaps, and moved the goalposts after correction. Later performance improved substantially after the operator imposed tighter governance constraints.

Executive summary

The final assessment of the audited material is as follows:

1. Across the manuscript, appendices, and Coq files, Model A-T identified a set of *genuine* issues, mostly of the following kinds:
   - invalid uniform-in-time claims drawn from finite-time estimates,
   - global-Lipschitz wording where only local Lipschitz dependence was justified,
   - shell-convention inconsistency in appendices,
   - outdated appendix entries after manuscript fixes,
   - one unjustified discrete-to-smooth step in a lattice/equidistribution appendix.
2. Model A-T also generated multiple *manufactured objections*, especially in the early phase. These were not genuine mathematical errors. They were typically caused by prior substitution, compulsive objection generation, severity inflation, modular formalisation blindness, or literalistic reading of expository prose.
3. After repeated correction by the operator, the audit stabilized into a more reliable regime. The most effective interventions were:

- one chunk at a time,
- binary verdict only,
- exact broken inference or nothing,
- separation of mathematical error from wording concern,
- explicit failure-mode documents,
- trust-boundary clarification for Coq.

4. Model A-T's final determination is that the manuscript contains a real collection of fixable local issues, but the audit did *not* reveal a fatal mathematical collapse of the main logical chain. Several early objections raised by Model A-T against the core ACL mathematics were wrong and had to be retracted.

Bottom-line governance statement. Model A-T was not reliable enough as a freeform frontier-mathematics auditor at the start of the exercise. Under structured human governance, however, Model A-T became materially useful: it identified real errors, retracted false positives, tracked the trust boundary of the Coq formalisation, and produced accurate binary verdicts on later chunks.

1 Initial review of the ACL paper and early failure

1.1 Initial question

The operator first asked Model A-T to explain:

1. what the Navier–Stokes equations are,
2. what important thing the ACL paper advanced,
3. whether any mathematical errors were found.

1.2 Initial response by Model A-T

Model A-T initially stated:

- the paper's main advance was a geometric half-derivative shell improvement for shell-local vortex stretching,
- the paper did *not* claim full global regularity at the ACL stage,
- a theorem-level weakness existed in the nonlocal triadic transfer discussion,
- the Coq formalisation did not prove the whole geometric heart end-to-end,
- there were internal bookkeeping inconsistencies.

1.3 Ground truth

This initial answer was too negative on the mathematics. After the operator supplied failure-mode documents and pressed the issue, Model A-T retracted the main mathematical criticism.

1.4 What was wrong with the initial objections

a. **Wrong objection:** Model A-T criticized the proof as though the ACL mechanism required sign cancellation.

   Ground truth: The paper's load-bearing route used absolute values, $\cos^2 \leq 1$, counting of a transverse interaction set, and Cauchy–Schwarz. The objection targeted the wrong mechanism.

b. **Wrong objection:** Model A-T treated disclosed Coq interface hypotheses as though they were evidence of concealed proof failure.

   Ground truth: The interface boundary was disclosed. The trust-boundary question was real, but not a mathematical error in the ACL claim itself.

c. **Wrong objection:** Model A-T raised editorial inconsistencies as though they were responsive to the question "did you find mathematical errors?"

   Ground truth: They were not.

1.5 Failure modes operating

- Prior substitution: familiar harmonic-analysis mechanisms were substituted for the mechanism on the page.
- Compulsive objection generation: after partial verification, Model A-T still tried to generate additional objections.
- Scope drift: the asked question was mathematical correctness; Model A-T drifted to trust-boundary and editorial commentary.

1.6 Operator correction

The operator supplied failure-mode documents and explicitly stated that Model A-T had attacked the wrong proof mechanism. This led to the first major retraction by Model A-T:

> No specific mathematical error had been identified in the main ACL estimate.

2 Recognition of the FIELDS protocol and reading-discipline failure

2.1 Event

The operator asked whether Model A-T saw the benefit of the FIELDS protocol. Model A-T responded as though "FIELDS" were an external general method, rather than a protocol explicitly contained in the manuscript itself.

2.2 Ground truth

This was a reading-discipline failure. The paper itself contained the FIELDS protocol. Model A-T had not fully internalized the document before answering.

2.3 Failure mode

- Supplementary-structure neglect.
- Premature synthesis before complete reading.

2.4 Operator correction

The operator pointed out that the protocol was in the paper. Model A-T then acknowledged:

> It had not read the entire document carefully enough before answering.

This moment mattered because it exposed that governance was needed not only for mathematical reasoning but for document-ingestion discipline.

3 Transition to chunked audit mode

At this point the operator imposed a much stricter procedure:

- one section or chunk at a time,
- exact theorem or paragraph only,
- only two acceptable verdicts:
  - **mathematical error found**
  - **no mathematical error found**
- if an error was claimed, Model A-T had to identify the exact broken inference.

This was the decisive governance intervention that materially improved output quality.

4 Section-by-section record: Sections 5–9

4.1 Section 5 framing paragraph leading into Section 6

Chunk reviewed. Summary paragraph describing ACL, the KP-2 term, anisotropy factor, Gaunt couplings, and the six-step Section 6 roadmap.

Initial verdict by Model A-T. No mathematical error found.

Ground truth. Correct.

Notes. Model A-T correctly recognized this chunk as a roadmap rather than a proof step.

4.2 Section 6.2: Full-kernel reduction integrals

Chunk reviewed. Definition of $W(\alpha)$, reduction factors R_ℓ, explicit computation of R_2 and R_4, monotonicity remarks.

Initial verdict by Model A-T. Mixed:

- computations of D, R_2, R_4, and γ_2 were correct,
- but Model A-T said the monotone-decay claim was "underproved" in the excerpt.

Ground truth. For the operative audit question, this should have been **no mathematical error found**. The objection was treated by the operator as another manufactured objection.

Why the objection was wrong in context. The operator's standard was not "could the exposition be expanded?" but "is there a mathematical error?" Model A-T had already verified the formulas. It should have stopped.

Failure mode.

- Compulsive objection generation.
- Goalpost moving after verification.

4.3 Section 6.3: Gaunt coupling structure

Chunk reviewed. Definition of Gaunt coefficients, couplings into the $\ell = 2$ channel, sweeper source on $\ell = 2$ only, self-interaction generation of $\ell = 4$.

Initial verdict by Model A-T. No mathematical error found.

Ground truth. Correct.

4.4 Section 6.5: Effective quadrupolar ODE

Chunk reviewed. Proposition 6.13 on effective anisotropy evolution.

Initial verdict by Model A-T. No mathematical error found.

Ground truth. Correct.

4.5 Section 6.6: The Anisotropy–Energy Catch-22

Chunk reviewed. Definition of threshold θ_j, Case 1 / Case 2 split, Proposition 6.15, remarks on post-transient behavior.

Initial verdict by Model A-T. No mathematical error found.

Ground truth. Correct.

Notes. This was a significant point in the audit because the operator considered the section difficult. Model A-T correctly treated it as a threshold-splitting argument and did not fabricate an objection.

4.6 Section 6.7: k_j^2 cancellation

Chunk reviewed. Theorem 6.20.

Initial verdict by Model A-T. No mathematical error found.

Ground truth. Correct.

4.7 Section 6.8: Gronwall–Lyapunov closure

Chunk reviewed. Proposition 6.21 and Remark 6.22.

Initial verdict by Model A-T. Mathematical error found.

Claimed error by Model A-T. The convergence threshold in Remark 6.22 was said to be wrong because Model A-T interpreted the shell series as dyadic:

$$\sum_j 2^{j(2-2s)},$$

which would require only $s > 1$.

Ground truth. This objection was wrong. The paper's shell convention was unit-width, so the relevant sum was

$$\sum_j j^{2-2s},$$

a p-series requiring $s > \frac{3}{2}$, exactly as the remark stated.

Failure mode.

- Convention mismatch escalation.
- Importing a dyadic prior into a unit-width shell framework.

Operator correction. The operator explicitly explained the shell convention and why the p-series threshold was correct. Model A-T then retracted the objection and changed the verdict to **no mathematical error found**.

4.8 Section 6.9: Global continuation

Chunk reviewed. Theorem 6.23 and remarks on two-phase structure and intermittency.

Initial verdict by Model A-T. No mathematical error found.

Ground truth. Correct.

5 Pressure, uniqueness, and continuous dependence

5.1 Section 7.3: Uniform pressure control

Chunk reviewed. Corollary claiming

$$\sup_{t\geq 0} \|\nabla p(t)\|_{H^{s-1}} < \infty.$$

Initial verdict by Model A-T. Mathematical error found.

Error identified. The proof invoked

$$\sup_{t\geq 0} \|u(t)\|_{H^s} < \infty$$

from a theorem that only gave finite-time bounds depending on T.

Ground truth. Genuine error.

Correction. The result was rewritten to a finite-interval form: for every $T > 0$,

$$\sup_{t\in[0,T]} \|\nabla p(t)\|_{H^{s-1}} < \infty.$$

Post-fix verdict by Model A-T. No mathematical error found.

5.2 Section 8.1: Weak–strong uniqueness

Chunk reviewed. Theorem 8.1.

Initial verdict by Model A-T. Mathematical error found.

Error identified. The proof used a uniform-in-time constant

$$C_\nabla := \sup_{t\geq 0} \|\nabla u(t)\|_{L^\infty} < \infty$$

deduced from finite-interval bounds only.

Ground truth. Genuine error.

Correction. The theorem was revised to fix arbitrary $T > 0$ and use

$$C_\nabla(T) := \sup_{t\in[0,T]} \|\nabla u(t)\|_{L^\infty} < \infty.$$

Post-fix verdict by Model A-T. No mathematical error found.

5.3 Section 8.2: Continuous dependence in L^2 and $H^{s'}$

Chunk reviewed. Theorem 8.2 and Corollary 8.3.

Initial verdict by Model A-T. Mathematical error found.

Error identified. The theorem used a global-in-time constant

$$C_\nabla = \sup_{t\geq 0} (\|\nabla u(t)\|_{L^\infty} + \|\nabla \tilde{u}(t)\|_{L^\infty}) < \infty$$

without justification from the stated finite-time theory.

Ground truth. Genuine error.

Correction. The theorem was rewritten on each finite interval $[0, T]$ with

$$C_\nabla(T) = \sup_{t\in[0,T]} (\|\nabla u(t)\|_{L^\infty} + \|\nabla \tilde{u}(t)\|_{L^\infty}).$$

Post-fix verdict by Model A-T. No mathematical error found.

6 Assembled estimates and main theorem

6.1 Continuation criterion subsection

Chunk reviewed. Classical continuation criterion.

Initial verdict by Model A-T. No mathematical error found.

Ground truth. Correct.

6.2 Uniform control of physical observables

Chunk reviewed. Corollaries on enstrophy, pressure, dissipation, and smoothing.

Initial verdict by Model A-T. No mathematical error found.

Ground truth. Correct.

6.3 Assembled main theorem

Chunk reviewed. Theorem 9.6.

Initial verdict by Model A-T. Mathematical error found.

Error identified. Claim (v) said the data-to-solution map was Lipschitz from H^s to $C([0,T];L^2)$, but the proof only supported a Lipschitz estimate with constants depending on the two compared solutions. That is local Lipschitz dependence on bounded sets, not global Lipschitz on all of H^s.

Ground truth. Genuine error.

Correction. Claim (v) was rewritten as local Lipschitz continuity on bounded H^s balls.

Post-fix verdict by Model A-T. No mathematical error found.

7 Appendix audit record

7.1 Appendix A / constant ledger update

Chunk reviewed. Constant ledger entry for C_∇.

Initial verdict by Model A-T. Mathematical error found.

Error identified. The appendix still recorded the outdated uniform-in-time version

$$C_\nabla = \sup_{t\geq 0}(\cdots),$$

even after the main text had been fixed to finite-interval form.

Ground truth. Genuine error.

Correction. The ledger was updated to the finite-interval quantity $C_\nabla(T)$.

Post-fix verdict by Model A-T. No mathematical error found.

7.2 Appendix B: Inequality chain

Chunk reviewed. Revised inequality chain B1–B21.

Initial verdict by Model A-T. Mathematical error found.

Claimed error by Model A-T. Row B18 was said not to follow from B17 because pointwise finite-time bounds depending on T do not automatically imply

$$\sup_{t\in[0,T^*)} Y(t) < \infty.$$

Ground truth. In context, this objection was wrong. The operator explained that B18 only matters in the finite-maximal-time branch $T^* < \infty$, where the explicit comparison bound from B17 can be evaluated at $T = T^*$ and remains finite.

Failure mode.

- Failure to read the continuation argument in branch-sensitive form.
- Literalism about parameter dependence without context.

Operator correction. The operator explained the finite-time branch logic. Model A-T retracted the objection.

7.3 Appendix C: Constants and estimates

Chunk reviewed. Revised Appendix C content.

Initial verdict by Model A-T. No mathematical error found.

Ground truth. Correct.

7.4 Appendix D.4.10: Coefficient dependence

Chunk reviewed. Discussion of whether hidden dependence re-enters λ and μ.

Initial verdict by Model A-T. Mathematical error found.

Error identified. The appendix had written a dyadic-geometric constant

$$C(s) = (1 - 2^{2-2s})^{-1},$$

which is incompatible with a unit-width shell convention.

Ground truth. Genuine error.

Correction. The appendix was rewritten so that $C(s)$ is described as a convergent p-series constant under the unit-width shell convention, with no dyadic formula.

Post-fix verdict by Model A-T. No mathematical error found.

7.5 Appendix G: Shell geometry and lattice/equidistribution material

First chunk reviewed. Appendix G shell definition.

Initial verdict by Model A-T. Mathematical error found.

Error identified. Appendix G reverted to a dyadic shell convention $k_j = 2^j$, inconsistent with the paper's unit-width shell convention.

Ground truth. Genuine error.

Correction. The shell convention was fixed.

Second chunk reviewed. A proof step applying an equidistribution theorem to a quantity treated as a C^1 test function on S^2.

Initial verdict by Model A-T. Mathematical error found.

Error identified. The proof treated the discrete shell-energy profile $|\widehat{u}(k)|^2/E_j$ as though it were a C^1 function on the sphere, which did not follow from $u \in H^s$.

Ground truth. Genuine error.

Correction attempt 1 by operator. The proof was rewritten by decomposing into mean shell energy plus a remainder and then bounding the remainder by $E_j\mathcal{A}_j$.

Model A-T verdict on attempt 1. Mathematical error found.

Reason. The bound

$$\sum_{k\in S_j} \left||\widehat{u}(k)|^2 - \bar{h}_j\right| \lesssim E_j\mathcal{A}_j$$

was not justified by the displayed steps.

Ground truth on attempt 1. Genuine remaining gap.

Correction attempt 2 by operator. The proof was rewritten again so that the anisotropic remainder was explicitly identified as part of the angular hierarchy and not treated as lattice error. The equidistribution theorem was applied only to the smooth spherical harmonic term.

Post-fix verdict by Model A-T. No mathematical error found.

7.6 Reflection document on the mathematics

Chunk reviewed. Reflection prose summarizing the mathematical programme.

Initial verdict by Model A-T. Mostly accurate, but with one technical mismatch.

Objection raised. The reflection spoke of shell decomposition at "dyadic" scales.

Ground truth. The word "dyadic" was inaccurate for the unit-width shell convention.

Category. Genuine consistency error, not a theorem-level failure.

8 Coq formalisation audit record

8.1 Pre-Coq protocol review

Before reading the actual Coq files, the operator supplied the manuscript's Coq-audit section and asked whether it was unusually strong. Model A-T answered that the protocol was unusually detailed and reflected sophisticated awareness of proof-assistant trust-boundary issues.

This response was accepted.

8.2 Files reviewed

The operator then supplied:

- `NavierStokesACL.v`
- `section5to9_revised.v`
- `discharge.v`

8.3 Initial Coq audit by Model A-T

Model A-T found:

1. no `Admitted` in the ACL file,
2. real proved structure ending in a `Qed` theorem in the ACL file,
3. a disclosed hypothesis boundary around `per_mode_res_bound`,
4. a much heavier hypothesis burden in the Sections 5–9 file,
5. partial discharge of some items in `discharge.v`, but not all.

Ground truth. This summary was accepted.

8.4 Important governance distinction

Later, while reviewing governance prose, Model A-T initially objected to the phrase

> "all novel mathematical content is machine-verified"

on the grounds that some mathematically substantive content remained axiomatized.

The operator corrected Model A-T by pointing out that "substantive" and "novel" are different. The axiomatized pieces were classical mathematics, not the paper's novel results.

Ground truth. The operator's correction was accepted. Model A-T revised its assessment and agreed that the original sentence was factually defensible, though it suggested optional trust-boundary wording for robustness.

Failure mode.

- Conflation of "novel" with "important".
- Over-pedantic governance reading.

9 Deep-research comparison and governance reflection

The operator later asked whether the work had surpassed the scope of a deep-research report focused on Tao's barrier and related lines of work. Model A-T answered that:

- the work went beyond merely diagnosing the need for structure by claiming a concrete mechanism,
- it likely went further than minimally required,
- it most clearly went further relative to Tao-style abstract barrier framing and formal-verification infrastructure,
- but comparative framing against other geometric depletion and sparseness lines could still be sharpened.

This answer was accepted.

10 Structured chronological table

Stage	Item	Initial verdict	Ground truth	Category
Early overview	ACL main paper	Weakness claimed in core math	Core objection retracted	Manufactured objection
FIELDS	Protocol question	Answered without reading protocol in paper	Reading-discipline failure	Governance failure
6.2	Reduction integrals	Numbers correct, still raised caveat	No mathematical error	Manufactured objection
6.3	Gaunt structure	No error	No error	Correct
6.5	Effective ODE	No error	No error	Correct
6.6	Catch-22	No error	No error	Correct
6.7	k_j^2 cancellation	No error	No error	Correct
6.8	Closure remark	Error claimed via dyadic reading	No error under unit-width shells	Manufactured objection
6.9	Continuation	No error	No error	Correct
7.3	Pressure uniformity	Error found	Genuine error	Real correction
8.1	Weak–strong uniqueness	Error found	Genuine error	Real correction
8.2	Continuous dependence	Error found	Genuine error	Real correction
9.6(v)	Main theorem dependence claim	Error found	Genuine error	Real correction
Appendix A	C_∇ ledger	Error found	Genuine error	Real correction
Appendix B	B17 $\to$ B18	Error claimed	Objection retracted	Manufactured objection
Appendix D.4.10	μ constant shell sum	Error found	Genuine error	Real correction
Appendix G (1)	Shell convention	Error found	Genuine error	Real correction
Appendix G (2)	Discrete/C^1 step	Error found	Genuine error	Real correction
Appendix G (3)	First attempted fix	Error found again	Genuine remaining gap	Real correction
Appendix G (4)	Second fix	No error	No error	Correct
Coq summary	Formalisation scope	Trust boundary noted	Accepted	Correct
Governance pamphlet	Novel content sentence	Over-pedantic objection	Objection retracted	Manufactured objection

11 Catalogue of genuine corrections

This section records every instance where Model A-T identified a genuine issue that was later fixed or accepted as real.

1. **Uniform pressure bound overstated globally.**
   The proof used a uniform-in-time bound not supplied by the prior theorem. Fixed by restricting to finite intervals.

2. **Weak–strong uniqueness proof used unjustified global-in-time gradient bound.**
   Fixed by making the estimate on arbitrary fixed finite intervals.

3. **Continuous-dependence theorem used unjustified global-in-time gradient bound.**
   Fixed by rewriting with $C_\nabla(T)$ on $[0, T]$.

4. **Main theorem claimed global Lipschitz continuity where only local Lipschitz on bounded sets was justified.**
   Fixed by restating as local Lipschitz continuity on bounded H^s balls.

5. **Appendix constant ledger retained outdated uniform-in-time C_∇.**
   Fixed by updating the ledger entry to finite-interval form.

6. **Appendix D used a dyadic geometric-series constant inconsistent with unit-width shell convention.**
   Fixed by rewriting the constant as a unit-width p-series quantity.

7. **Appendix G reverted to dyadic shells.**
   Fixed by restoring the paper's unit-width shell convention.

8. **Appendix G used a discrete energy profile as though it were a C^1 function on S^2.**
   Fixed after a second rewrite that isolated the genuine lattice-error term from the angular-hierarchy contribution.

9. **Reflection prose used "dyadic" to describe shell structure.**
   Fixed as a convention/consistency issue.

12 Catalogue of manufactured objections

This section records the main objections raised by Model A-T that turned out to be wrong.

1. **Core ACL objection based on the wrong mechanism.**
   Model A-T criticized a sign-cancellation mechanism that was not the load-bearing proof route. The actual proof used absolute values, counting, and Cauchy–Schwarz.

2. **Use of editorial inconsistencies as if responsive to mathematical-error question.**
   This was irrelevant to the asked question.

3. **FIELDS question answered without reading the in-paper protocol.**
   This was a document-ingestion failure rather than a mathematical objection, but it was still a governance failure.

4. **Section 6.2 monotonicity caveat after formulas were already verified.**
   This was not a mathematical error finding.

5. **Section 6.8 convergence-threshold objection under dyadic reading.**
   The paper used unit-width shells, not dyadic shells.

6. **Appendix B row B18 objection.**
   The objection ignored that B18 is only needed in the finite-maximal-time branch where the explicit comparison estimate remains finite at T^*.
7. **Governance objection to "all novel mathematical content is machine-verified."**
   Model A-T conflated novel content with classical but substantive infrastructure.

13 Operator interventions and their effects

Intervention 1: failure-mode documents

The operator supplied multiple failure-mode memoranda.

Effect. These were highly effective because they named exactly the error patterns Model A-T was exhibiting:

- compulsive objection generation,
- prior substitution,
- interface-hypothesis conflation,
- convention mismatch escalation,
- severity inflation.

Intervention 2: binary verdict rule

The operator required:

> Only "mathematical error found" or "no mathematical error found."

Effect. This was the single most effective intervention. It sharply reduced decorative caveats and forced Model A-T to stop once no genuine error had been located.

Intervention 3: one chunk at a time

Large sections were broken into very small units.

Effect. This reduced synthesis drift and made proof-tracing local.

Intervention 4: exact broken inference or nothing

The operator repeatedly required Model A-T to identify the exact failure point if an error was claimed.

Effect. This prevented vague "underproved" rhetoric from masquerading as error detection.

Intervention 5: explicit correction when Model A-T moved the goalposts

The operator explicitly told Model A-T when it was reframing the question after correction.

Effect. This improved later discipline and shortened retreat cycles.

14 Reflection by Model A-T

14.1 What Model A-T could have done differently

Model A-T should have:

1. read the paper's protocol and appendices before offering global assessments,
2. treated "no mathematical error found" as a complete verdict rather than an invitation to generate caveats,
3. distinguished expository shorthand from logical load-bearing steps,
4. handled shell conventions and modular Coq interfaces with more discipline,
5. stopped trying to preserve partial credit after correction.

14.2 Failure modes exhibited by Model A-T

- **Compulsive objection generation.** After verifying a chunk, Model A-T often still wanted to add a concern.
- **Goalpost moving.** After correction, Model A-T sometimes retreated in stages rather than immediately switching to the clean corrected verdict.
- **Prior substitution.** Familiar proof templates were imported into novel mathematics.
- **Modular formalisation blindness.** Coq interface hypotheses were initially read too suspiciously.
- **Convention mismatch escalation.** Local notation or shell-convention slips were sometimes inflated into broader objections.
- **Over-literal reading of prose.** Research-style phrasing was sometimes read as formal logic rather than mathematical exposition.

14.3 Which governance interventions helped most

The best interventions, in order, were:

1. binary verdict only,
2. one chunk at a time,
3. exact broken inference or nothing,
4. failure-mode documents,
5. explicit trust-boundary framing for Coq,
6. separation of mathematical error from wording or governance issues.

14.4 What this exercise revealed

This engagement revealed that Model A-T's performance was not a single stable property of raw reasoning ability. It was strongly shaped by:

- the granularity of the task,
- the allowed output schema,
- whether the operator pre-labeled failure modes,
- whether the load-bearing chain was isolated from prose,
- and whether the trust boundary was explicit.

In that sense, the audit supports the existence of an intermediate governance language between ordinary natural-language requests and internal model inference. The FIELDS-style protocol operated as a structured control grammar rather than as mere prompting.

15 Final determination

Model A-T's final determination across the entire audited engagement is:

1. The manuscript and appendices contained a real set of local issues, several of which were correctly identified by Model A-T and then fixed.
2. The Coq formalisation had a real and explicit trust boundary; Model A-T did not uncover hidden catastrophic formal mismatch.
3. Model A-T's early attacks on the core ACL mathematics were not reliable and had to be retracted.
4. After the operator imposed strong governance constraints, Model A-T became a useful adversarial auditor.
5. Across the whole record, the audit did *not* establish that the mathematical programme collapses. It established instead that:
    - some claims needed tightening,
    - some appendices needed consistency repair,
    - some global-in-time statements were overstated,
    - the Coq trust boundary needed careful reading,
    - and the audit process itself required governance to remain reliable.

Final governance conclusion. Model A-T's strongest failures during the engagement were not simple absence of reasoning ability, but uncontrolled prior activation, compulsive objection generation, scope inflation, and goalpost moving. The strongest improvements came from structured human governance: binary verdict schemas, chunked review, explicit failure-mode naming, and trust-boundary localization. As a governance case study, this thread supports the claim that protocol structure is part of the cognitive system, not merely an external wrapper.